85th PERCENTILE 24-HR STORM SIMULATION

The 85th percentile, 24-hour storm is an important design storm for land development and water quality planning in Los Angeles County (County). To support regional applications, WMMS 2.0 boundary conditions (".pre" and ".air" files) have been generated to simulate the 85th percentile, 24-hour storm. As characterized by the Los Angeles County Department of Public Works (LACDPW 2004), the 85th percentile 24-hour storm is the depth of precipitation determined from a spatially distributed statistical analysis using rainfall stations within the County. The most recent isohyets resulting from this spatial statistical analysis are available from the County GIS Portal (LAC GIS Portal 2016). These data were applied to the LSPC model subwatersheds by area-weighted average to calculate a depth for each. Figure 1 shows the resulting 85th percentile 24-hr storm depth for all subwatersheds in the model domain. For subwatersheds in neighboring counties (Ventura, Orange, San Bernardino), the same statistical analysis developed by LACPDW was applied to the historical rainfall record (water year 1991 to 2018). Reference the *WMMS 2.0 Phase 1 Report: Model Calibration and Validation* for a description of how this timeseries was developed (available in WMMS 2.0 Repository).

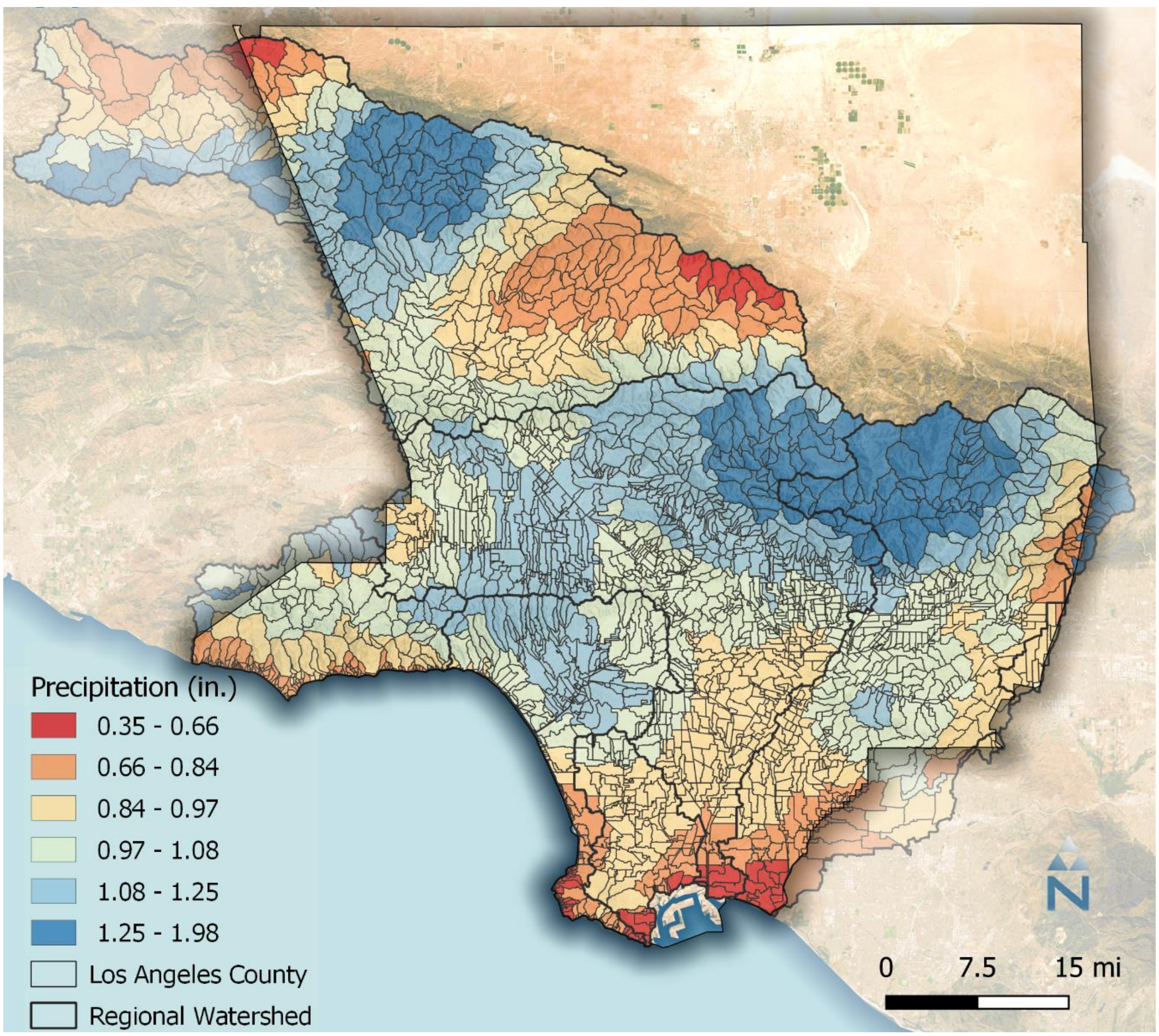

Figure 1. 85th percentile 24-hr storm depth across the WMMS 2.0 model domain.

Temporal distribution of the 85th percentile storm depth used the LACDPW 4-day unit hyetograph for typical storm banding (LACDPW 2006; see Figure 2). The unit hyetograph was scaled to each subwatershed's precipitation depth and formatted as LSPC precipitation timeseries files available for download on the WMMS 2.0 Repository. Use of a 4-day storm is conservative for runoff volume because soils in the simulation would be fully saturated by the fourth day and therefore can be used for assessing runoff volume reduction. Because it is not conservative for water quality (pollutants have not had sufficient time to build-up), this timeseries is not recommended for assessing water quality reduction.

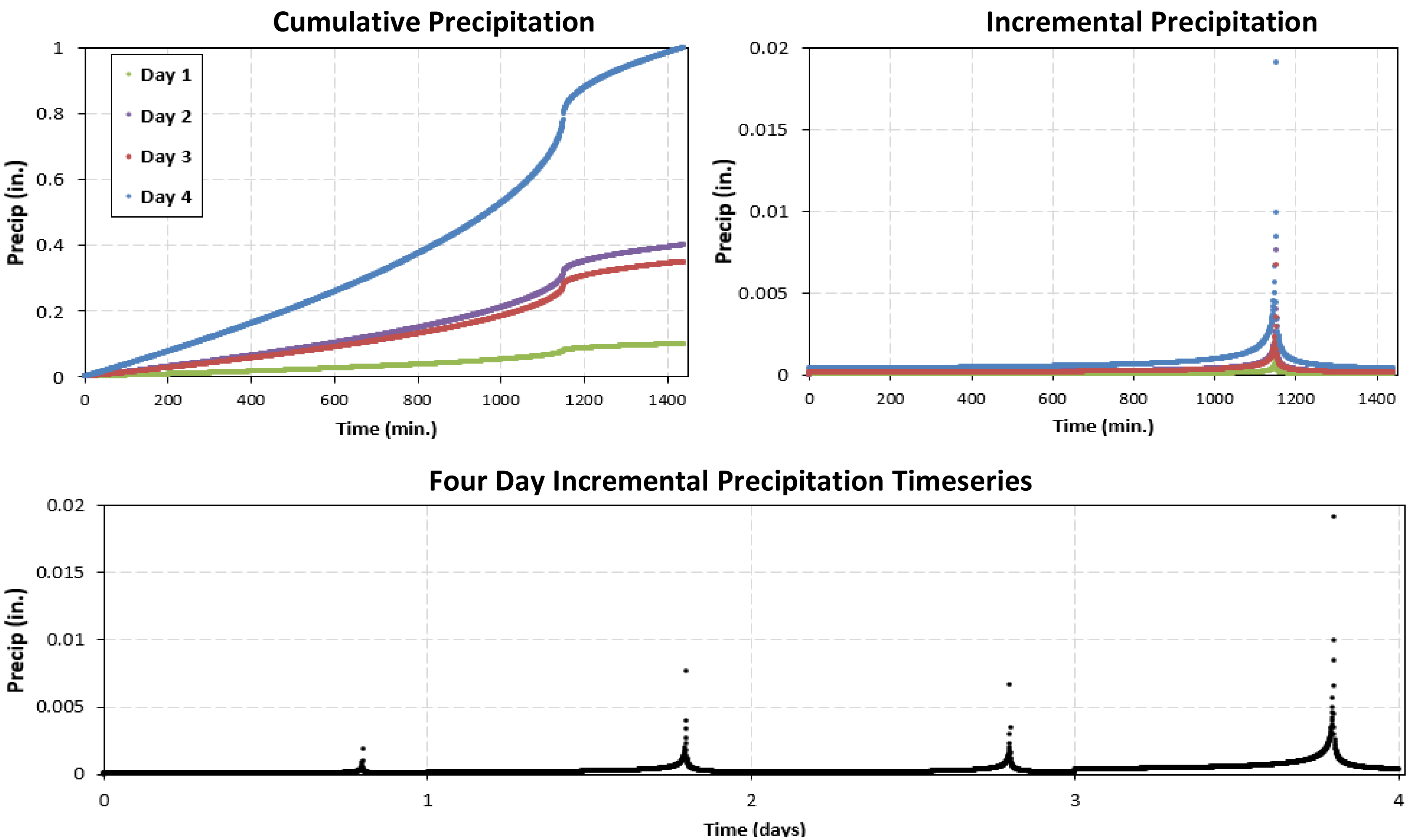

Figure 2. Cumulative (top left) and incremental precipitation (top right and bottom) for the 1-in. unit storm using the LACDPW 4-day design storm hyetograph.

REFERENCES

LACDPW (Los Angeles County Department of Public Works). 2004. Analysis of 85th Percentile 24-hour Rainfall Depth Analysis Within the County of Los Angeles. Water Resources Division. February 2004. Retrieved January 2020 from https://www.ladpw.org/wrd/publication/engineering/Final_Report-Probability_Analysis_of_85th_Percentile_24-hr_Rainfall1.pdf

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